Going Deeper: Why Depth Works in CNNs

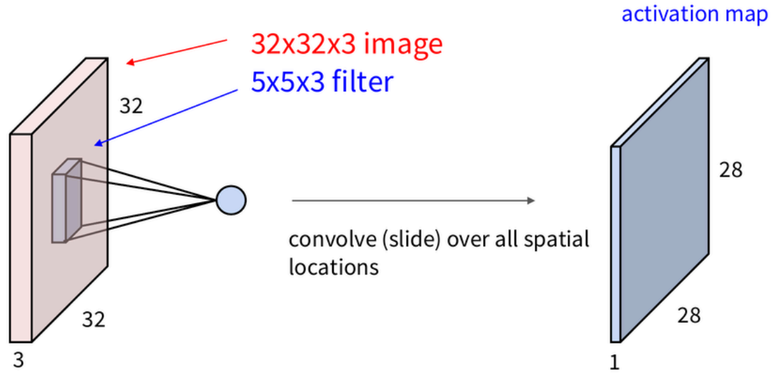
How three networks (AlexNet, VGG, ResNet) made it possible

Application No. [MT21021] · Junhyeong Lee

DGIST Artificial Intelligence Major · M.S. Admission Interview · October 21, 2026

The core idea: convolution = a sliding dot product

A filter slides over the image: one dot product per position



$$w^T x + b$$

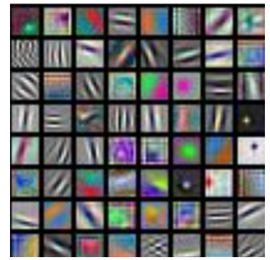
large where the patch looks like the filter

Why not a fully-connected layer?

- 224×224×3 image → 150k inputs → 150M weights for 1,000 units
- Every pixel connects to every unit → position is ignored



FC learns: blurry whole-image templates



Conv learns: local edges & colors

Why does convolution fit images?

① Local connectivity

Each unit sees only a small patch → position is kept

② Weight sharing

Same filter everywhere → few weights, found anywhere

③ Stacking

Edges → parts → objects

④ Learned

Filter values are weights, trained by backpropagation

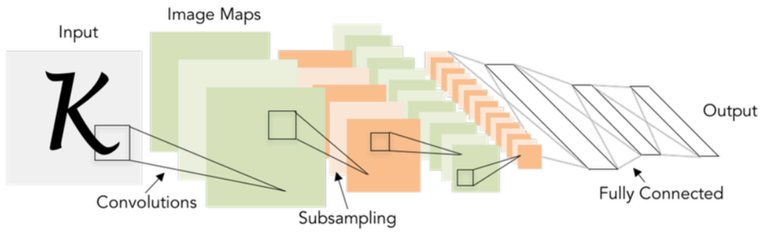
Deeper layers see more of the image and build richer features. → how deep can we go?

Going deeper: LeNet → AlexNet → VGG

LeNet-5 1998

5 learnable layers · MNIST digits

- conv + pooling + FC
- Too little data and compute to go deeper

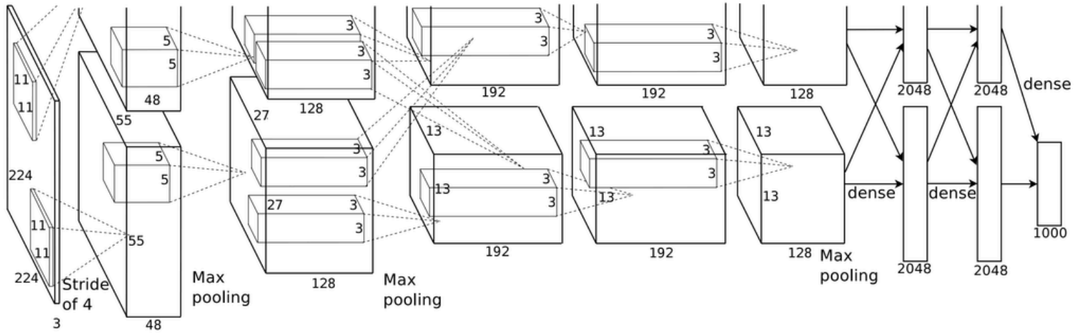


LeCun et al., 1998

AlexNet 2012

8 layers · 60M params · ImageNet top-5 error 25.8% → 16.4%

- GPUs + 1.2M images, ReLU, dropout, augmentation
- 5 conv + 3 FC, split over two GPUs

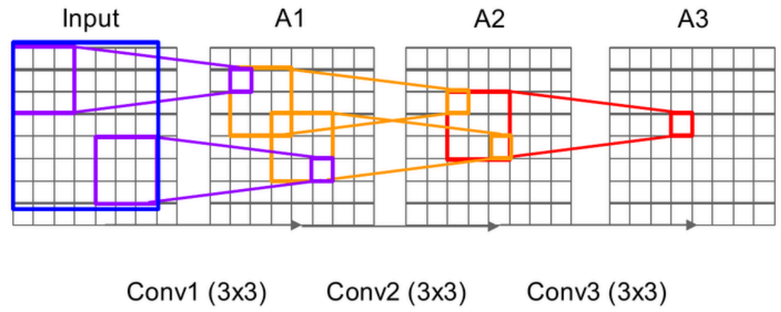


Krizhevsky et al., 2012

VGG 2014

16–19 layers · top-5 error 7.3%

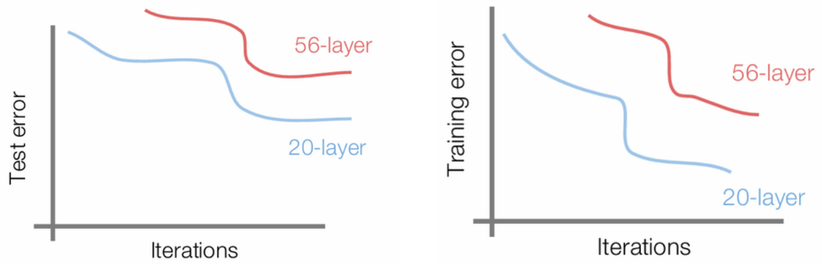
- Only 3×3 filters: three 3×3 = one 7×7 receptive field, $27C^2$ vs $49C^2$ weights (C = channels), more non-linearity
- Same simple block, repeated → just go deeper



AlexNet made depth trainable. VGG made depth simple. Lesson: deeper is better — if you can train it

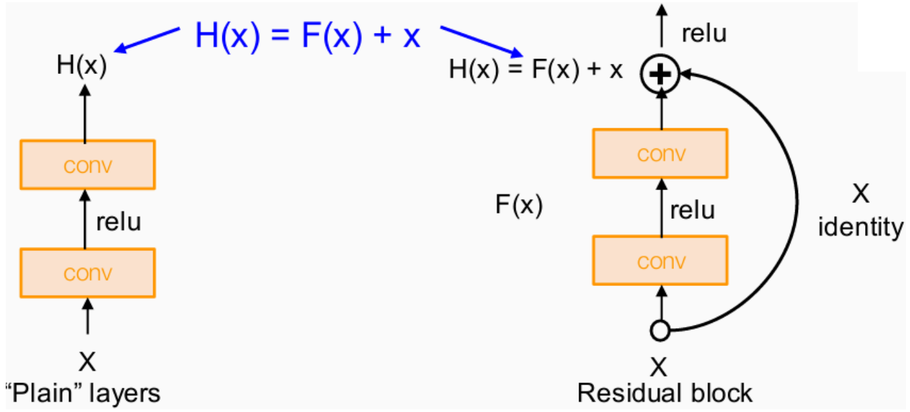
Deeper isn't free: the degradation problem and ResNet

Problem: 56 plain layers do worse than 20



- Even training error is worse
  → not overfitting. Deeper models are harder to optimize.

Solution: the residual block

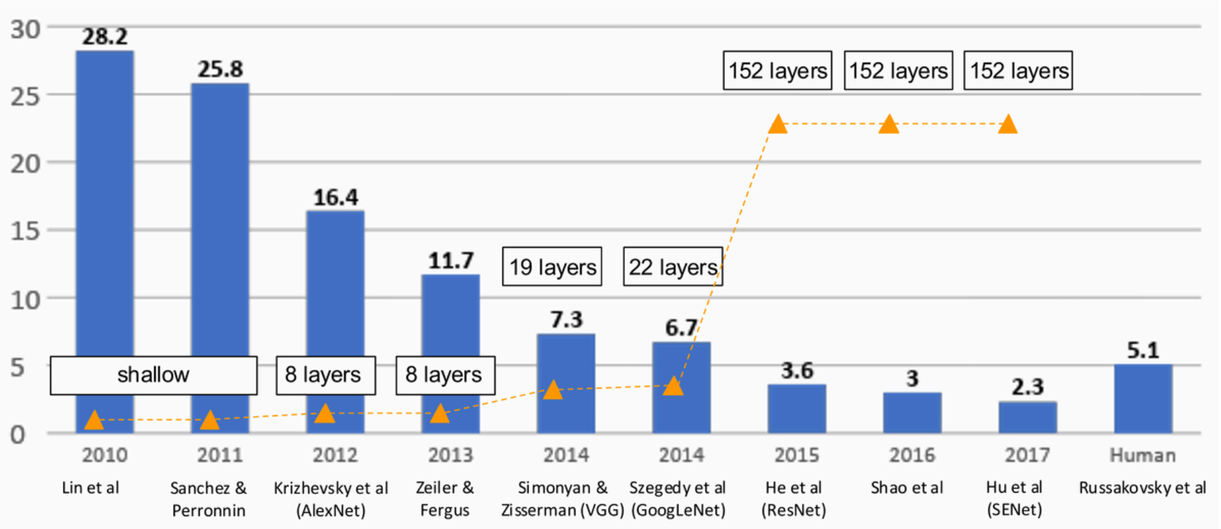


He et al., 2016

- Layers learn only F(x) = H(x) − x → F = 0 gives identity
- Gradient flows straight through the skip path

Result: 152 layers, error below human level

ImageNet top-5 error (%)



	AlexNet	VGG	ResNet	Human
Layers	8	19	152	-
Top-5 error	16.4%	7.3%	3.6%	5.1%

What made depth possible was the skip connection, not just more layers.

Summary: what three networks taught us about depth

AlexNet · 2012

made depth TRAINABLE

GPUs, 1.2M images, ReLU, dropout
→ 8 layers, 16.4% error

VGG · 2014

made depth SIMPLE

Only 3×3 filters: three 3×3 = one 7×7, fewer weights, more non-linearity
→ 19 layers, 7.3%

ResNet · 2015

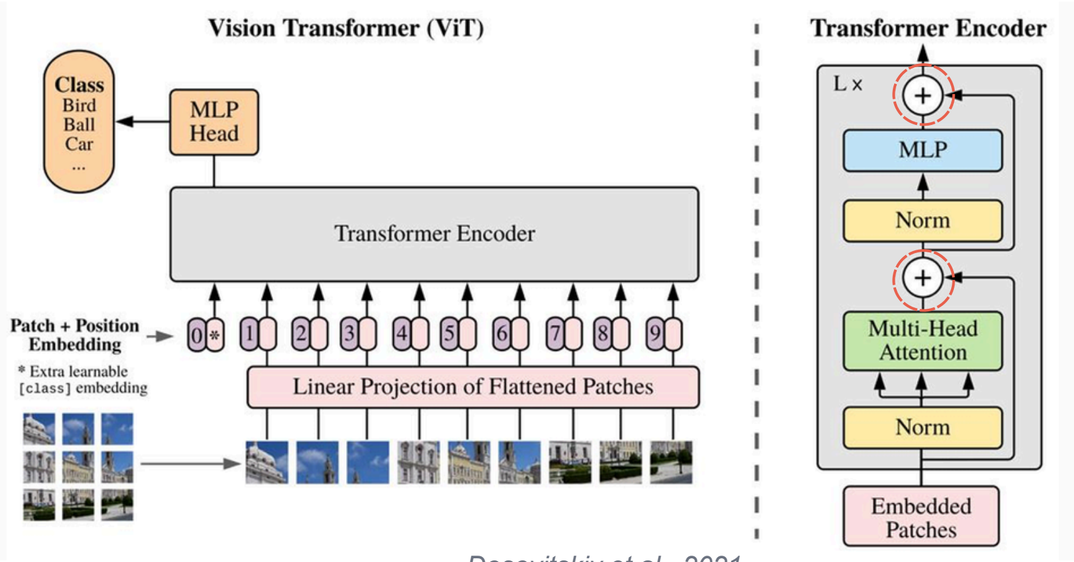
made depth OPTIMIZABLE

Skip connection: learn F(x) = H(x) − x, identity is easy to learn
→ 152 layers, 3.6%

Depth is what makes deep learning powerful — residual connections are what made deeper networks trainable.

Beyond CNNs

- Transformers (2017) and Vision Transformers (2020) keep the **residual path** for exactly the same reason: it makes deep stacks trainable.
- ConvNeXt (2022): a modern CNN with residual blocks still matches Transformers



Dosovitskiy et al., 2021

Figures adapted from Stanford CS231n (Lectures 5–6) and the original papers.